

Gray-Zone Maritime Control in the First Island Chain

A networked escalation game

Workshop 1

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USS Chafee (DDG 90). U.S. Navy photo

Background: why we built this wargame

Strategic context

- The South China Sea is a major maritime trade route and a contested strategic region.
- The United States conducts naval operations to uphold navigational rights and challenge excessive maritime claims.
- China seeks to turn its maritime claims into sustained control, with persistent coast guard and militia presence backed by navy ships.
- Much of this competition takes place below the threshold of war.
- **Recent example.** On October 20, China reported monitoring USS Charleston near Sabina Shoal and accused it of entering without authorization. The U.S. described the activity as routine operations in international waters (Reuters).
- The game captures the resulting tradeoff between presence, control and escalation. Players allocate ships and choose whether to patrol or assert. Stronger presence can improve local control, while confrontational encounters can produce incidents that raise tension across the theater. The model simplifies this regional competition into two players, Blue and Red.

Why a wargame

Each side's best move depends on the other's, and the risk of war is shared. Real encounters cannot be rerun or varied on purpose, so a wargame lets us test strategies and force designs without real risk.

What models exist

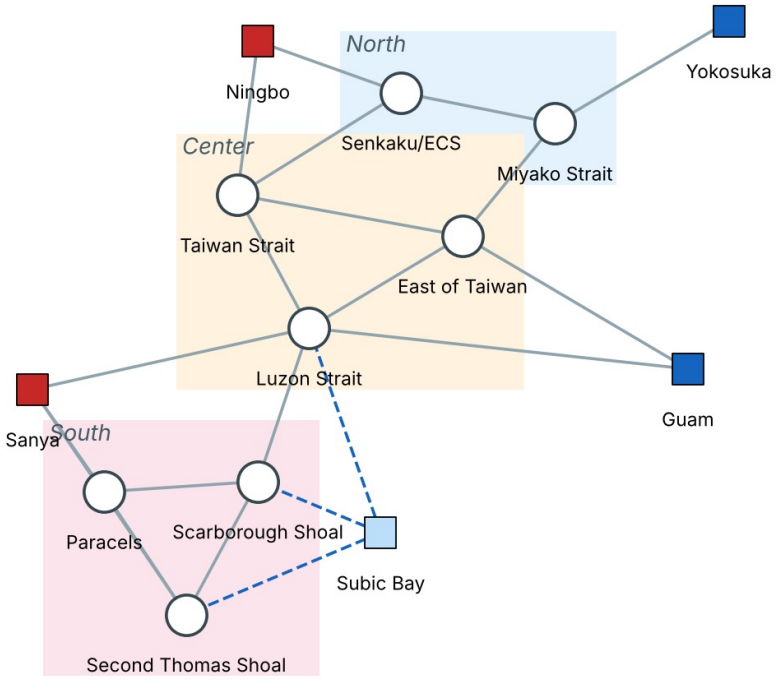
Gray-zone games with probabilistic adjudication (Wasser 2019), randomized patrols (Shieh 2012), network interdiction (Washburn 1995), and escalation as a shared risk (Schelling 1966).

What we add

Maritime allocation, evolving control and shared escalation risk in one general-sum game, with computational rules so we can compare strategies and designs across thousands of games.

Takeaway Each existing model captures part of this contest. Our game combines them and runs thousands of times.

The problem: control without war



Network of 8 contested zones and 4 home ports

- **Players.** Blue: 10 U.S. destroyers (DDG). Red: 8 navy ships (PLAN) + 6 coast guard / militia hulls (CCG).
- **Each period (2 days).** Each side chooses where its ships go and whether to patrol (P) or assert (A). Asserting means shadowing, close maneuvering, or querying and warning. For the U.S. it also covers freedom-of-navigation operations.
- **Control** of a zone follows the local force ratio, so each side's best move depends on the other's.
- **Incidents** can happen when both sides are present, and are likelier if someone asserts. Incidents raise an escalation level. At the top is war, which costs both sides.
- **Fog.** Neither side sees the other's entire fleet disposition, only forces present where they also maintain forces. Storms can close zones.

Research questions and what changed since the proposal

RQ1

Control vs. war risk

Each side caps how much incident risk it will accept each period (its risk budget ε). How much more control does a side gain by accepting more risk, how much more likely does war become, and what risk level does each side pick when it knows the other side is picking too?

RQ2

Game theory vs. rules of thumb

Does a side that plans with game theory (best responses, equilibrium or maximin plans) do better than one that follows simple rules or moves at random? Can both sides keep patrolling over many periods because asserting once would end the truce and push both into riskier play from then on?

RQ3

Weather and redesign

Does the plan or Blue redesign that works best in normal weather still work best across calm, typhoon and extreme weather scenarios, both on average and in the worst scenario?

Feedback	What we did
Model too high level	Formal game on a graph
Q1 is a feasible region, use a MIP	Best response is an exact MIP. Commander types become risk budgets ε
More ambitious questions	Budget-game equilibrium (RQ1), equilibrium play (RQ2), robustness and redesign (RQ3)
Define the resilient redesign	Three Blue designs
Propose values for missing data	Assumed values, listed in the report
Show preliminary results	Analytical results, plus 77 setups × 200 simulated games
Add weather (Prof. Sapsis)	Weather scenarios, scored by expected and worst-case payoff

The game, one period at a time

1. The weather (sea state) of each region is drawn
2. Each side sees its own ships and detects some enemy ships
3. Ships low on endurance head home
4. Both sides choose moves and postures at the same time
5. Moves happen within movement budgets
6. Control updates. Incidents are drawn. The escalation level moves
7. Payoffs are paid. The game stops at war or after 60 periods

Control of zone v

$$c_v^{t+1} = (1-\lambda)\, c_v^t + \lambda\, \phi_v, \qquad \phi_v = \frac{S_{\mathrm{B},v} - S_{\mathrm{R},v}}{S_{\mathrm{B},v} + S_{\mathrm{R},v} + \kappa}$$

Presence builds control slowly. Leaving lets it decay.

Incident chance

$$h_v = \big(h_0 + h_{\mathrm{A}}^{\mathrm{B}}\, \mathbf{1}[p_{\mathrm{B},v} = \mathrm{A}] + h_{\mathrm{A},v}^{\mathrm{R}}\, \mathbf{1}[p_{\mathrm{R},v} = \mathrm{A}]\big) \cdot \min\{S_{\mathrm{B},v}, S_{\mathrm{R},v}\} \cdot \big(1 + \ell^t/2\big), \qquad \rho_v = 1 - e^{-h_v}$$

Needs both sides present. Grows with assertion and with the escalation level ℓ.

Stage payoff

$$g_i^t = \sigma_i \sum_{v \in \mathcal{C}} V_v^i\, c_v^{t+1} - \gamma\, m_i^t - \eta\, \ell^{t+1} - \zeta_D\, \mathbf{1}[i = \mathrm{B}]$$

Control value minus fuel, tension and Blue's redesign cost. Both pay W = 200 at war.

Takeaway Every rule is explicit, so the same game can be played thousands of times by different strategies.

Formulation: state, moves, escalation and payoffs

State
Ships by type and node, endurance, control, escalation level, sea state

$$s^t = (x^t, d^t, c^t, \ell^t, w^t), \quad x^t_{k,v} \in \mathbb{Z}_{\geq 0}, \quad d^t_j \leq D_k, \quad c^t_v \in [-1, 1], \quad \ell^t \in \{0, \ldots, L\}, \quad w^t_r \in \{\text{calm}, \text{rough}, \text{storm}\}$$

Flow conservation
No ships are added or removed

$$x^{t+1}_{k,u} = x^t_{k,u} + \sum_{v \in \mathcal{N}(u)} f^t_{k,vu} - \sum_{v \in \mathcal{N}(u)} f^t_{k,uv}$$

Movement budget
At most F_i ship moves per period (8 Blue, 10 Red)

$$\sum_{k \in \mathcal{K}_i} \sum_{(u,v)} f^t_{k,uv} \leq F_i$$

Local strength
Hulls weighted by type (CCG 0.6, 0.3 in rough seas), × α = 1.5 when asserting

$$S_{i,v} = \alpha^{\mathbf{1}[p_{i,v}=\text{A}]} \sum_{k \in \mathcal{K}_i} w_k \, x_{k,v}$$

Escalation
N^t = incidents this period. War when ℓ reaches L = 5

$$\ell^{t+1} = \begin{cases} \min\{L, \ \ell^t + N^t\} & \text{if } N^t \geq 1 \text{ incidents occur} \\ \max\{0, \ \ell^t - 1\} & \text{with probability } \mu \text{ if } N^t = 0 \\ \ell^t & \text{otherwise} \end{cases}$$

Total payoff
Discounted stage payoffs (δ = 0.98, T = 60), minus W = 200 for both sides at war

$$u_i = \sum_{t=1}^{\tau} \delta^{t-1} g^t_i - W \, \mathbf{1}[\ell^\tau = L], \qquad \tau = \min\{t : \ell^t = L\} \wedge T$$

Takeaway Both sides pay W at war, so the game is general-sum. Both prefer to avoid war.

How the players decide

Best response as a MIP

Each period, choose moves and postures to maximize value of control minus costs, subject to the movement budget, endurance, and an escalation-risk budget.

Expected incidents ≤ ε

Conservative / moderate / aggressive = ε of 0.05 / 0.15 / 0.40

Template game

Abstract each side to 14 templates. A template says how to split ships North / Center / South, and whether to patrol or assert.

A 14×14 strategic-form game

Solved with iterated elimination of strictly dominated strategies (IESDS), Nash equilibrium, maximin LP, correlated-equilibrium LP

Policies we compare

Random moves
Rule of thumb (rule-based)
Quick best response (greedy)
Best response (MIP)
Nash equilibrium plan (equilibrium)
Safety-first plan (maximin)

Same random draws and weather for every policy

Takeaway The proposal's strategy types are now one number, ε, inside an optimization (a MIP).

The best-response MIP with an escalation-risk budget

$$\max_{y,\,b} \quad \sum_{v\in\mathcal{C}}\sum_{(n,p)} a_v(n,p)\, b_{v,(n,p)} - \gamma \sum_{j}\sum_{w\neq \mathrm{loc}(j)} y_{j,w} - 1000\,\xi \qquad \text{control value} - \text{incident cost} - \text{fuel}$$

$$\text{s.t.} \quad \sum_{w} y_{j,w} = 1 \quad \forall j \qquad \text{one destination per free ship}$$

$$\sum_{j}\sum_{w\neq \mathrm{loc}(j)} y_{j,w} \le F_i - (\text{forced moves}) \qquad \text{movement budget}$$

$$\sum_{(n,p)} b_{v,(n,p)} = 1 \quad \forall v \qquad \text{one option per zone}$$

$$\sum_{(n,p)} n_k\, b_{v,(n,p)} = \bar{x}_{k,v} + \sum_{j \text{ of type } k} y_{j,v} \quad \forall v,k \qquad \text{option matches the moves}$$

$$\sum_{v\in\mathcal{C}}\sum_{(n,p)} \rho_v(n,p)\, b_{v,(n,p)} \le \varepsilon_i\,(1-\ell^t/L) + \xi \qquad \text{escalation-risk budget}$$

$$y,\, b \in \{0,1\}, \quad \xi \ge 0$$

- **Variables.** y = 1 sends free ship j to node w. b = 1 picks option (n, p) for zone v, meaning hull counts n and posture p.
- **Why options.** Control φ and incident risk ρ are nonlinear in ship counts, so each option's value and risk are computed in advance. The program stays linear.
- **Risk budget.** Expected incidents ≤ ε, shrinking to zero as ℓ nears war. ε = 0.05 / 0.15 / 0.40 is conservative / moderate / aggressive.
- **Slack ξ** covers unavoidable risk from forced returns, at a penalty of 1000.

Value of an option (control gain minus incident cost)

$$a_v(n,p) = V_v^i\,\beta\,\sigma_i\,\phi_v(n,p) - \Lambda\,\rho_v(n,p), \qquad \beta = \frac{\lambda}{1-\delta(1-\lambda)} \approx 0.96$$

Takeaway The risk budget ε defines the feasible region. It is how the proposal's commander types enter the model.

Small example: when does it pay to assert?

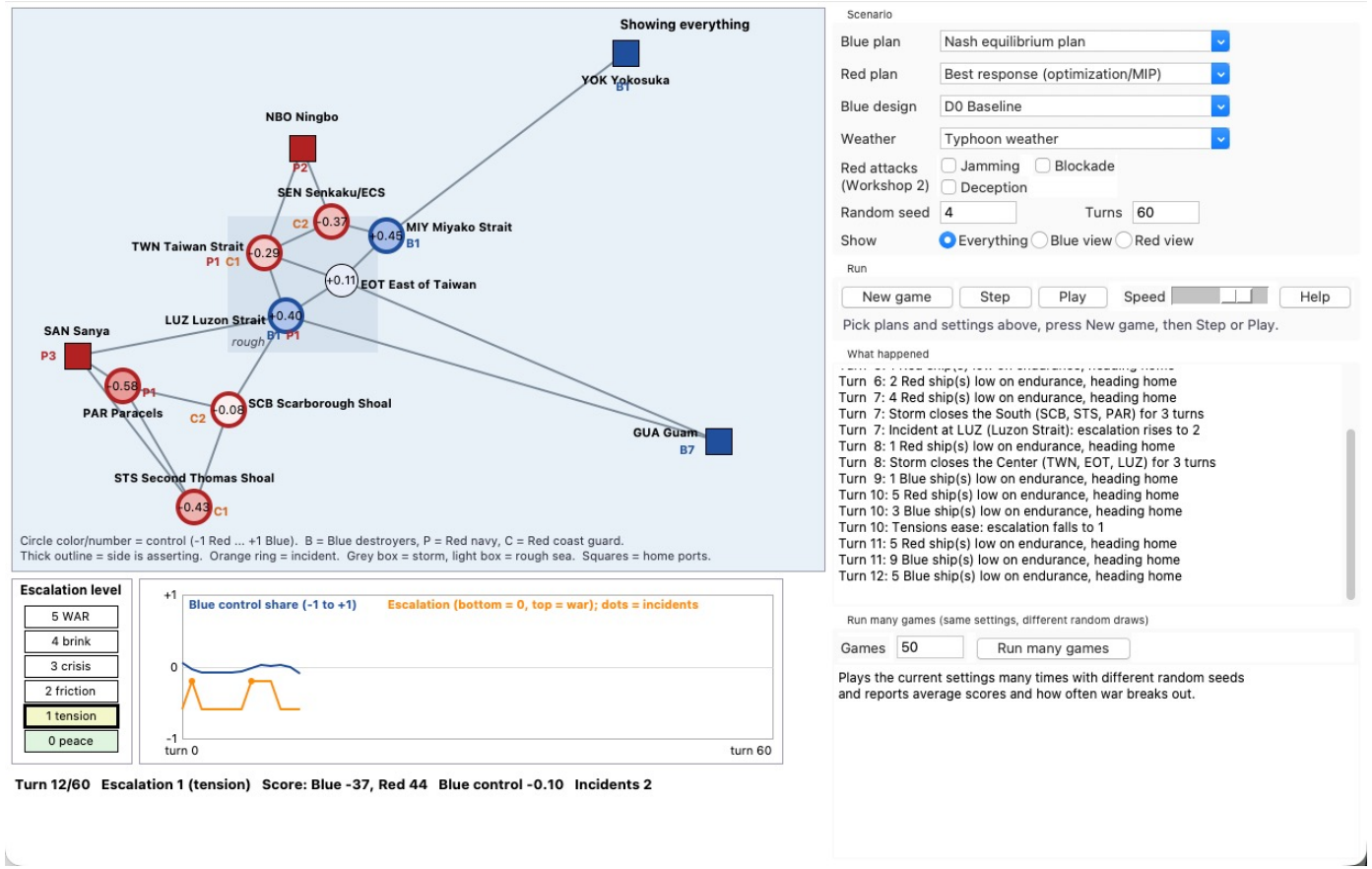
Blue \ Red	Red patrols	Red asserts
Blue patrols	0 , 0	−9.6 , +1.2
Blue asserts	+0.3 , −10.5	−11.4 , −11.3

Payoffs (Blue, Red) with force splits fixed, relative to both patrolling

- **Best reply:** assert if the other patrols, patrol if the other asserts. Two pure equilibria, plus a mixed one with mutual assertion 8% of the time and payoffs (−1.30, −6.02).
- **A mediator helps:** the best correlated equilibrium never recommends (assert, assert). A hotline can play that role.
- **Repetition helps:** if both patrol and any assertion makes both fall back to the mixed equilibrium for good, mutual patrolling is a subgame perfect equilibrium for any discount factor above 0.18. Ours is 0.98.

Takeaway Restraint is not stable in a one-shot encounter, but it can be sustained by repeated play or a coordination device.

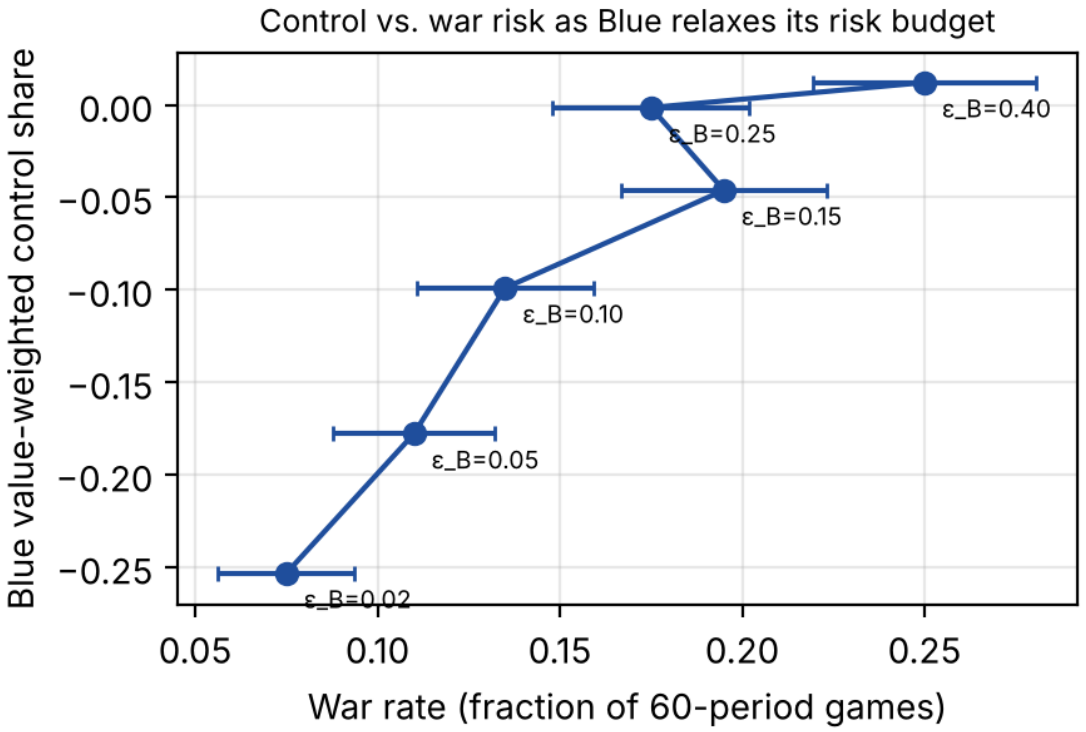
The prototype: watch a game play out



- **Pick** a plan for each side, a design and the weather on the right. Press New game, then Step or Play. Help explains the map.
- **Zone color** shows control. Blue = Blue, red = Red.
- **B / P / C** = destroyers, navy ships, coast guard. Thick outline = asserting.
- Orange ring = incident. Shaded region = rough sea or storm.
- **Run many games** plays the same setup as many simulated games and reports averages.

Program demo

RQ1: more risk buys control, up to a point



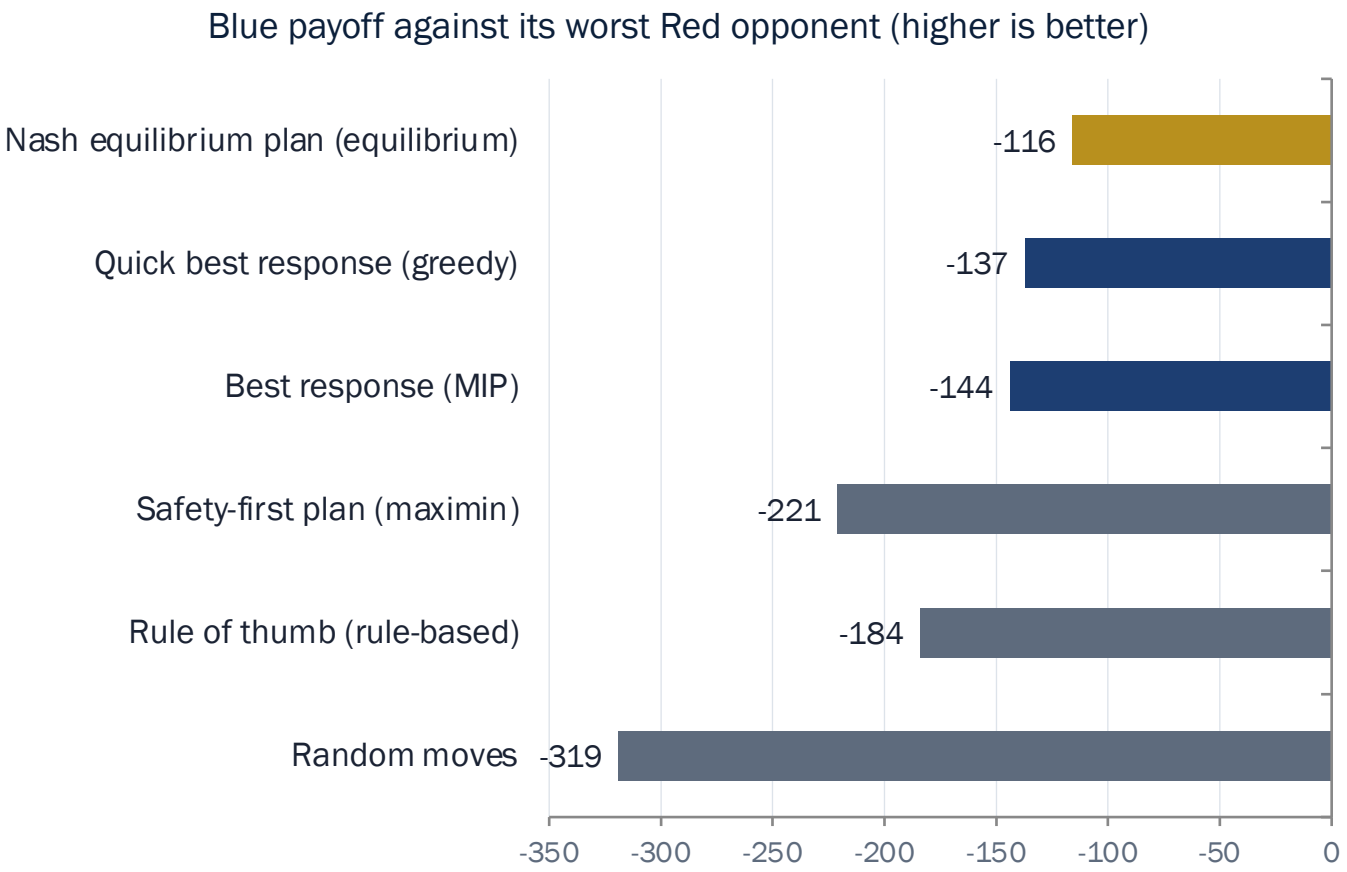
Blue \ Red	Cons. 0.05	Mod. 0.15	Aggr. 0.40
Cons. 0.05	−194, 199 war 10%	−303, 315 war 11%	−308, 318 war 12%
Mod. 0.15	18, −36 war 15%	−144, 101 war 20%	−192, 140 war 21%
Aggr. 0.40	74, −80 war 14%	−83, 16 war 25%	**−151, 39 war 34%**

Risk-budget game. Each side picks ε. Payoffs (Blue, Red) and war rate. Gold bold = the only equilibrium.

Raising Blue's budget from 0.02 to 0.15 lifts its control a lot. Beyond 0.25 it barely moves.

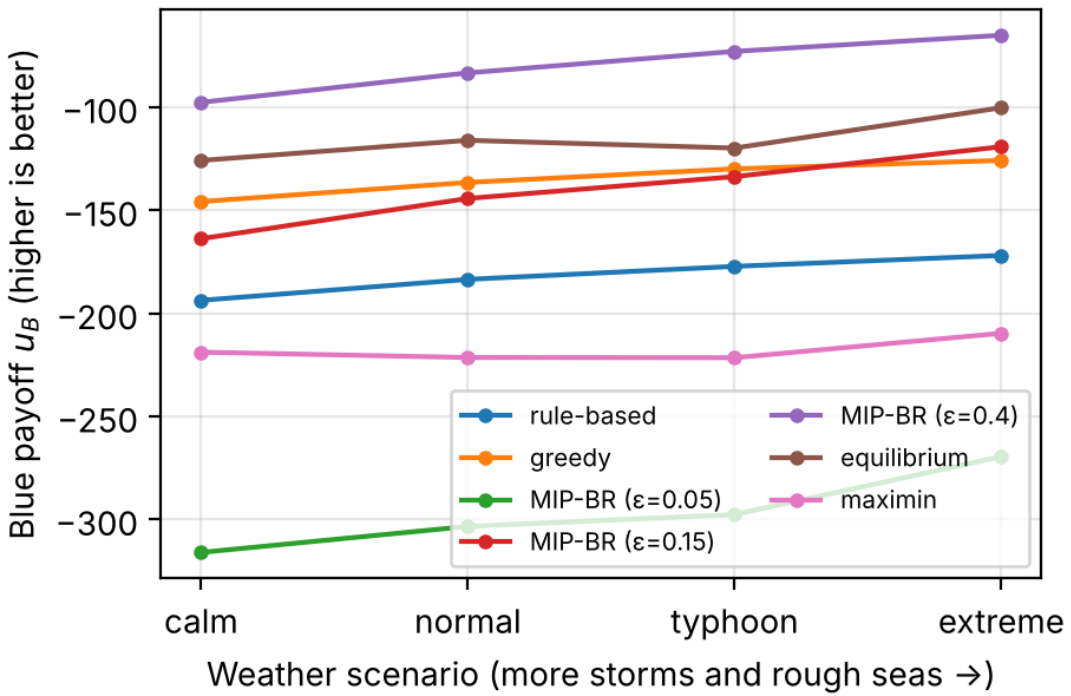
Takeaway Both choose aggressive (−151, 39), yet both do better at moderate (−144, 101). That is a prisoner's dilemma.

RQ2: the Nash equilibrium plan is the safest bet against any Red

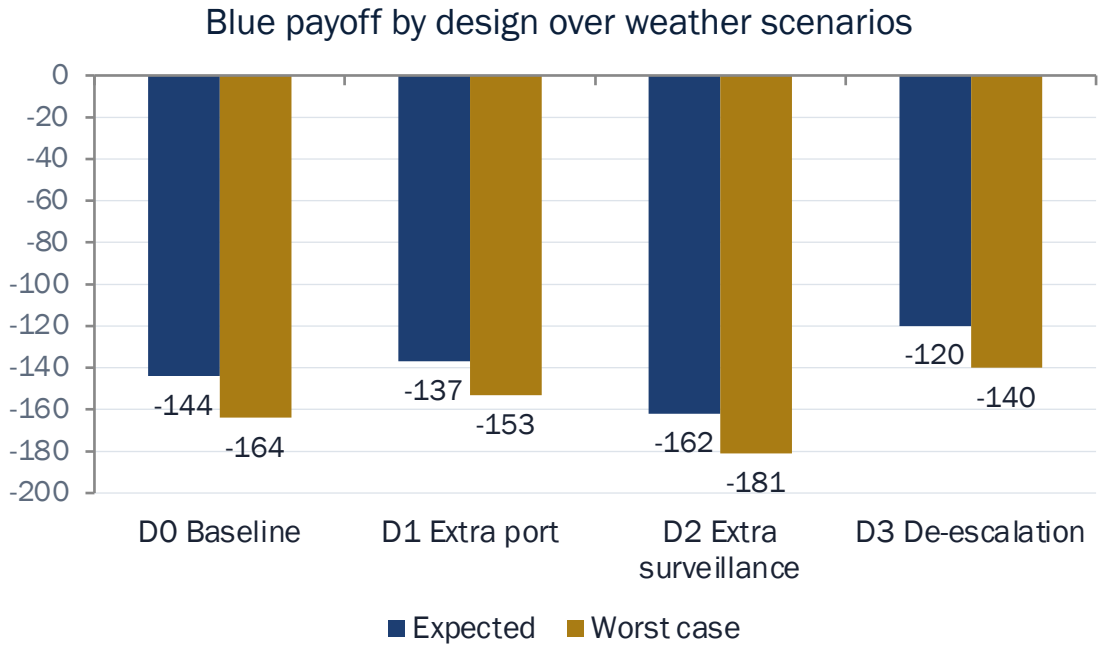


- Nash equilibrium plan has the best worst case (−116) and half the war rate of the MIP best response against the same opponent (10% vs 20%).
- Best responses (quick and MIP) win against weak opponents but escalate more.
- **Safety-first plan** is the most cautious (war ≤ 8%) but gives up the most control.
- **Quick vs MIP:** the quick heuristic matches the MIP 67% of the time but breaks the risk budget 14% of the time.

RQ3: the best strategy holds up in bad weather



Rough seas weaken Red's coast guard hulls and storms pause operations, so most policies do slightly worse in calm seas. The top two plans stay on top in every scenario.



War rate in normal weather is D0 20%, D1 28%, D2 23%, D3 5%. De-escalation measures were given no cost yet.

Takeaway The plan that is best on average is also best in the worst weather. De-escalation measures are the best design and cut war about 4×.

What the evidence does and does not show

Supported

- Loosening the risk budget has sharply diminishing returns
- Left alone, both sides loosen their budgets (prisoner's dilemma)
- Restraint can be an equilibrium of the repeated game
- The Nash equilibrium plan is the most robust Blue policy here
- De-escalation measures are a joint gain under every weather scenario

Not yet shown

- All numbers are assumptions, not calibrated to data
- Policies look only one period ahead
- War penalty may be too small. The top-payoff policy also has the most wars
- De-escalation measures have no cost yet, so part of their win is built in
- Differences under about 20 payoff points are ties

Plan for Workshop 2

Model

Beliefs, weather, budget

Bayesian beliefs and multi-period best response. Sensitivity analysis on V, h, W and μ. Calendar-based weather from real data. Reformulate the movement budget F_i with considerations for oilers.

Learning

Learning agents

A Red agent that learns from repeated play, and other learning methods. Do they converge, drift toward restraint, and exploit rule-based Blue?

Attacks

Attacks and defenses

Coast guard blockade at SCB and STS, jamming that halves Blue's detection in one region, and false contacts. A Red attacker that reacts to Blue's posture. Randomized patrols, D1 and D2 as defenses.

Designs

Resilient designs

D0 to D3 and combinations under attack and a long typhoon. A cost for D3.

Takeaway Workshop 2 adds better beliefs and weather, learning agents, attacks and defenses, and costed designs.